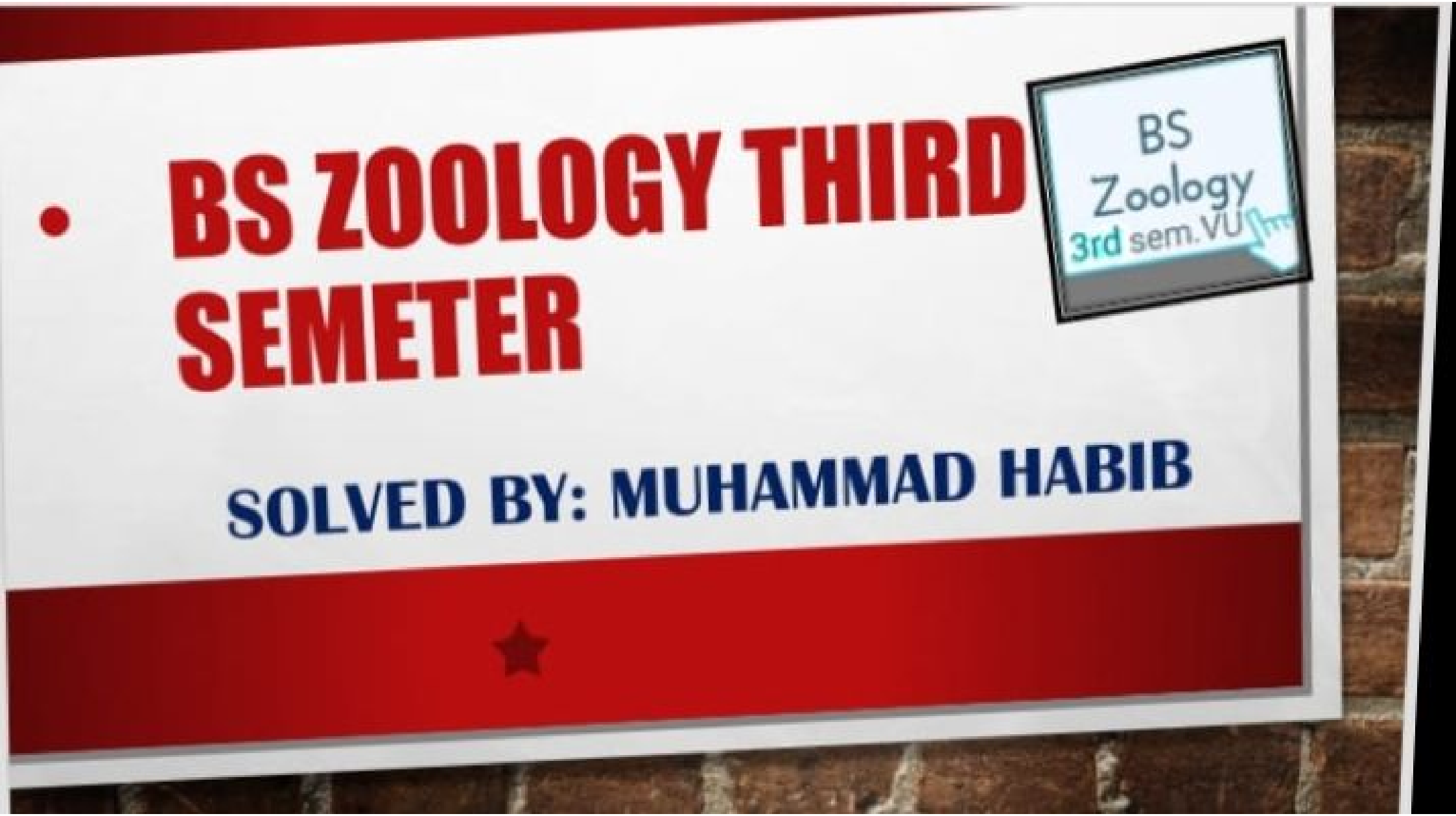

BS ZOOLOGY THIRD SEMETER
BS Zoology 3rd sem.VU
SOLVED BY: MUHAMMAD HABIB

Question # 10 of 10 (Start time: 03:42:45 PM, 13 August 2021)

A change in one nucleotide may alter the nucleotide sequence so that the ---------- fails to recognize and cleave at that site

Select the correct option

- ○ Polymerase I,II,III
- ○ Taq polymerase
- ○ Helicase
- ● Restriction endonuclease

Type here to search

5:08 PM

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quiz.vu.edu.pk/QuizQuestion.aspx?ver=e1e9cc12-d4ed-4604-a3fa-2a13f3f129a2

Apps Gmail YouTube Maps Home

Question # 9 of 10 (Start time: 03:42:07 PM, 13 August 2021)

For the detection and characterization of viral proteins, monoclonal and polyclonal antibody ——— technique is used

Select the correct option

- ○ Eastern blotting
- ● Western blotting
- ○ Northern blotting
- ○ Southern blotting

Type here to search

BIO203 Quiz 2

Question # 6 of 10 (Start time: 03:40:50 PM, 13 August 2021)

DNA Mutations can be identified by ----------

Select the correct option

- ○ Eastern blotting
- ○ Northern blotting
- ● Southern blotting
- ○ Western blotting

Type here to search

BIO203:Quiz 2

Question # 7 of 10 (Start time: 03:41:21 PM, 13 August 2021)

During western blotting, transfer can be done in

Select the correct option

- (●) Wet or semi-dry conditions
- ( ) Semi wet or semi-dry conditions
- ( ) Dry or semi-wet conditions
- ( ) Dry or semi-dry conditions

Type here to search

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BIO203 Quiz 2

Question # 8 of 10 (Start time: 03:41:44 PM, 13 August 2021)

Western blotting technique is simply a way to identify unknown ---------- on a polyacrylamide gel

Select the correct option

- Proteins
- Carbohydrate
- RNA
- DNA

Type here to search

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Apps Gmail YouTube Maps Home

BIO203:Quiz 2

Question # 5 of 10 (Start time: 03:39:44 PM, 13 August 2021)

The process of detection of the bands complementary to the probe, during Southern blotting is called ----------

Select the correct option

- ○ None
- ● Autoradiography
- ○ Biography
- ○ Autography

Type here to search

BIO203:Quiz 2

Question # 4 of 10 (Start time: 03:39:26 PM, 13 August 2021)

Which is not true for Radioactive Probe - p32

Select the correct option

- ○ Hazardous
- ○ Sensitive
- ● Radioactive waste disposal regulations should not be followed
- ○ Relatively cheap

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BIO203 Quiz 2

Question # 3 of 10 (Start time: 03:38:52 PM, 13 August 2021)

In Western blotting technique, incorrect labeling of the protein can happen due to the reaction of ---------- antibody

Select the correct option

- (●) Secondary
- ( ) Primary
- ( ) Tertiary
- ( ) None of above

Type here to search

8:17 ··· 83
https://quiz.vu.edu.pk/C 58

BC190408009: JAWERIA AQEEL

Time Left 78 sec(s)

BIO203:Quiz 2

Quiz Start Time: 08:11 PM, 13 August 2021

Question # 10 of 10 (Start time: 08:17:20 PM, 13 August 2021)

Total Marks: 1

Northern blot is a method used for detection of a specific ---

Select the correct option

- [x] RNA
- [ ] Proteins
- [ ] DNA
- [ ] Carbohydrate

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 03:37:16 PM, 13 August 2021)

Labeled material to detect complementary region in the gene or DNA is called as

Select the correct option

- DNTPs
- Probe
- Biotin
- Blocking agents

3:44 PM

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quiz.vu.edu.pk/QuizQuestion.aspx?ver=c29eeac2-91ac-4b5e-bd5c-f8653e7a9e9e

Apps Gmail YouTube Maps Home

uestion # 2 of 10 (Start time: 03:37:41 PM, 13 August 2021)

_____ is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein

Select the correct option

- Eastern blotting
- Western blotting
- Northern blotting
- Southern blotting

Type here to search

8:15 You...
30 photos
https://quiz.vu.edu.pk/C
58

BC190408009: JAWERIA AQEEL

Time Left 69 sec(s)

BIO203:Quiz 2

Quiz Start Time: 08:11 PM, 13 August 2021

Question # 8 of 10 (Start time: 08:15:02 PM, 13 August 2021)

Total Marks: 1

A change in one nucleotide may alter the nucleotide sequence so that the ----------fails to recognize and cleave at that site.

Select the correct option

- ○ Helicase
- ● Restriction endonuclease
- ○ Taq polymerase
- ○ Polymerase I,II,III

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 08:14:52 PM, 13 August 2021) Total Marks: 1

Western blotting technique is also used to detect

Select the correct option

- ○ Hepatitis-B
- ● All
- ○ Mad cow disease
- ○ Lyme Disease

Saving...

BC190408009: JAWERIA AQEEL

Time Left 57 sec(s)

BIO203:Quiz 2

Quiz Start Time: 08:11 PM, 13 August 2021

Question # 6 of 10 (Start time: 08:14:10 PM, 13 August 2021)

Total Marks: 1

Diagnosis of HIV by ELISA, involves the ---------- blotting technique.

Select the correct option

- ○ Northern
- ● Western
- ○ Southern
- ○ Eastern

Click to Save Answer & Move to Next Question

https://quiz.vu.edu.pk/C

Question # 2 of 10 (Start time: 08:12:05 PM, 13 August 2021)

Total Marks: 1

----------is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blotting membrane.

Select the correct option

- [x] Blocking
- [ ] Gel Electrophoresis
- [ ] Antibody probing
- [ ] Blotting or transfer

Click to Save Answer & Move to Next Question

Time Left sec(s)

BIO203:Quiz 2

Quiz Start Time: 08:11 PM, 13 August 2021

Question # 3 of 10 (Start time: 08:12:54 PM, 13 August 2021)

Total Marks: 1

DNA Mutations can be identified by ----------

Select the correct option

- Western blotting
- Eastern blotting
- Southern blotting
- Northern blotting

Click to Save Answer & Move to Next Question

https://quiz.vu.edu.pk/C

BC190408009: JAWERIA AQEEL

Time Left 78 sec(s)

BIO203:Quiz 2

Quiz Start Time: 08:11 PM, 13 August 2021

Question # 5 of 10 (Start time: 08:13:50 PM, 13 August 2021)

Total Marks: 1

---------- is sometimes called as protein blotting or immunoblotting

Select the correct option

- Northern blotting
- Eastern blotting
- Western blotting
- Southern blotting

Click to Save Answer & Move to Next Question

BC190408009: JAWERIA AQEEL

Time Left 78 sec(s)

BIO203:Quiz 2

Quiz Start Time: 08:11 PM, 13 August 2021

Question # 1 of 10 (Start time: 08:11:49 PM, 13 August 2021)

Total Marks: 1

In Western blotting technique, incorrect labeling of the protein can happen due to the reaction of ---------- antibody

Select the correct option

- ○ Primary
- ● Secondary
- ○ None of above
- ○ Tertiary

Click to Save Answer & Move to Next Question

Northern Blotting

PROCEDURE

- Isolation of intact mRNA.
- Separation of RNA according to size (through a denaturing agarose gel.
- Transfer of the RNA to a solid support.
- Fixation of the RNA.
- Hybridization of the immobilized RNA to probes complementary to the sequences of interest.

Regards : Zarva Chaudhary & Chaudhary Moazza

Transfer Methods - Southern

TRANSFER METHODS OF DNA TO MEMBRANE

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

Regards : Zarva Chaudhary & Chaudhary M

Transfer Methods - Southern

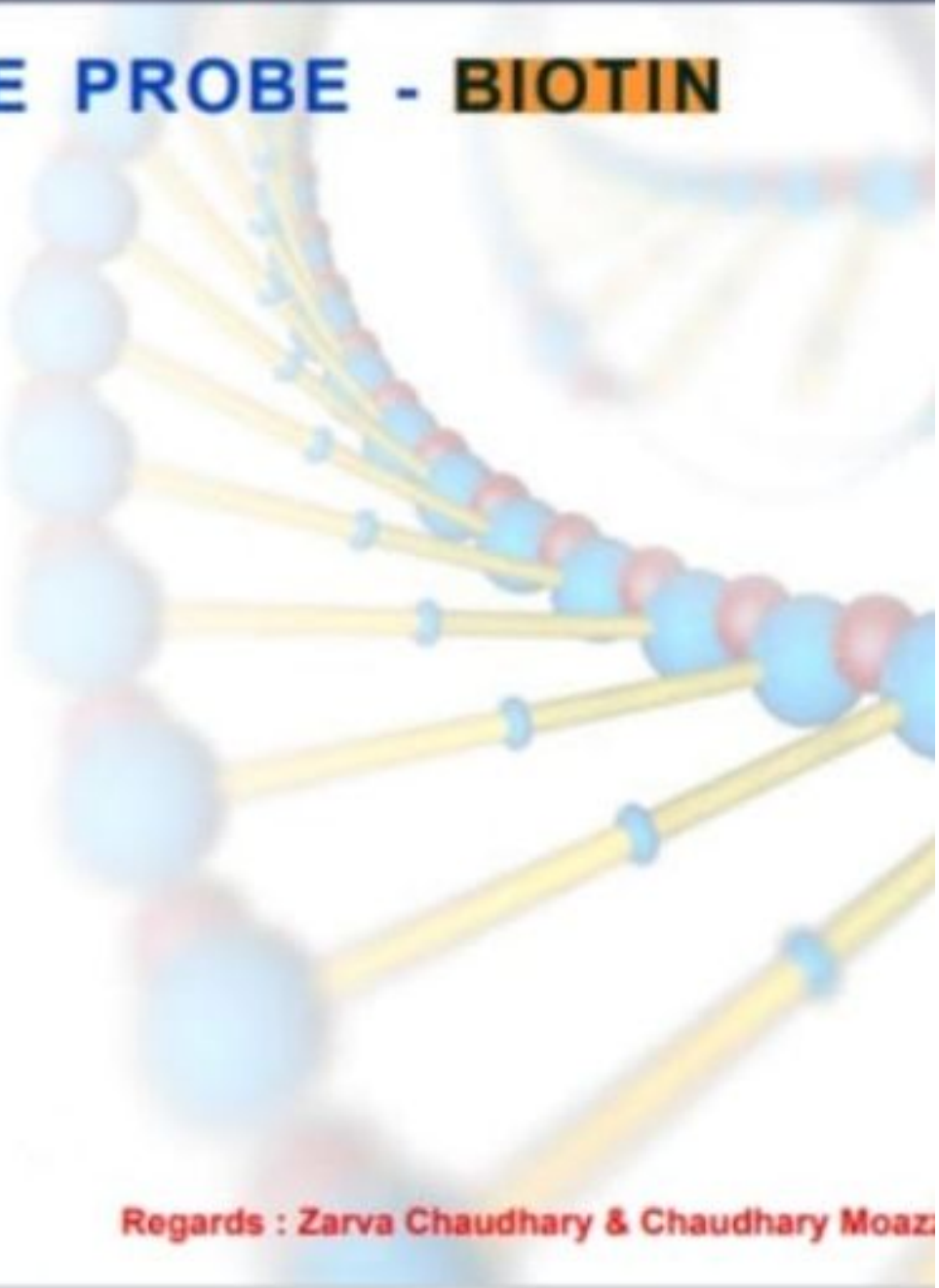
UPWARD TRANSFER



Southern Blotting - Probes

NON-RADIOACTIVE PROBE - BIOTIN

- Sensitive
- Relatively expensive

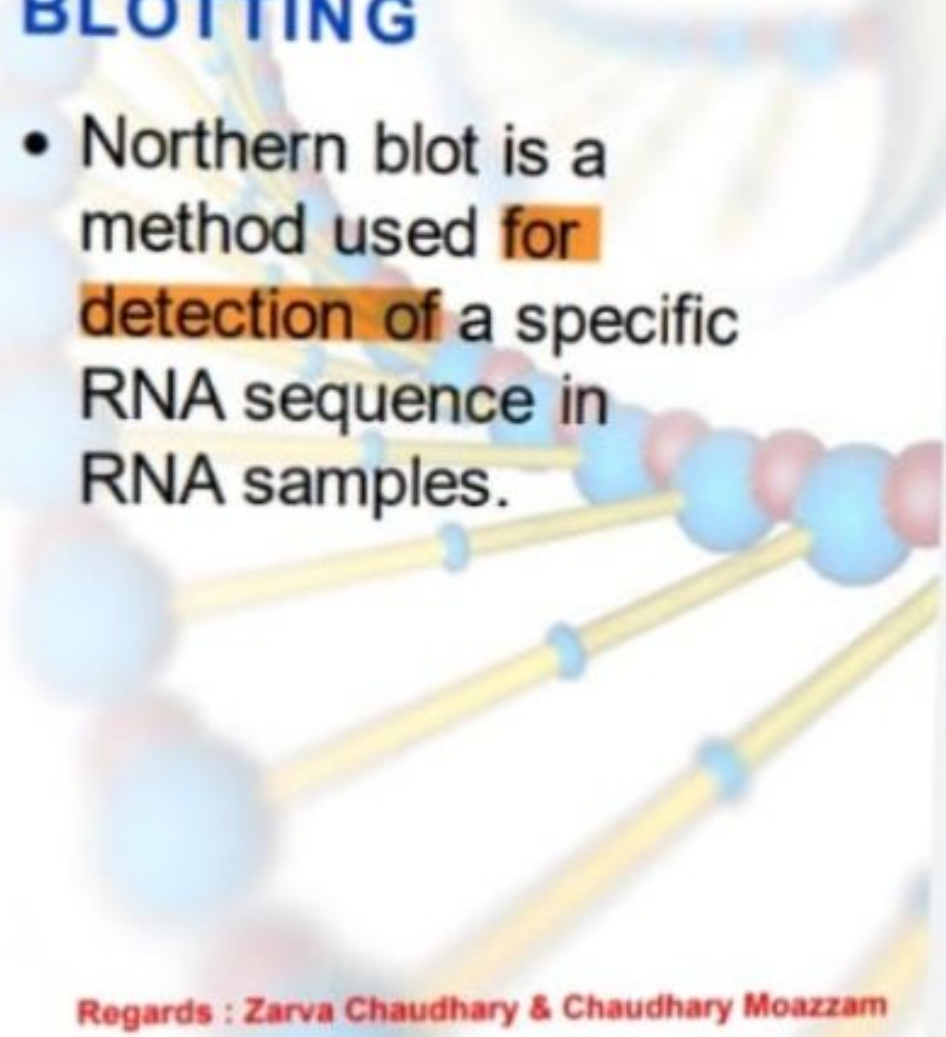


Regards : Zarva Chaudhary & Chaudhary Moaz

Northern Blotting

NORTHERN BLOTTING

- Northern blot is a method used for detection of a specific RNA sequence in RNA samples.



Regards : Zarva Chaudhary & Chaudhary Moazzam

Of Lyme Disease.

- Western blotting technique is used in Definitive Test For BSE which is commonly know as Mad cow disease.
- Confirmatory Test For Hepatitis-B involves western blotting technique.
- Western blotting test is used in the Analysis Of Biomarkers such as hormones, growth factors & cytokines.
- This technique is also employed in The Gene Expression Studies

Limitations in western blotting

- very delicate and time consuming process. A minute imbalan at any level of the procedure can skew the results of the entir process.
- Incorrect labeling of the protein can happen due to the reactio of secondary antibody.
- Cause erroneous in bands or no bands due to insufficient transfer.
- Well trained technicians are required for this technique.
- Primary antibody availability is crucial.
- It is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein is possible

transfer.

Well trained technicians are required for this technique.

Primary antibody availability is crucial.

It is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein is possible

Regards : Zarva Chaudhary & Chaudhary Mo

Conclusion

Western blotting technique is simply a way to identify unknown proteins on a polyacrylamide gel.

It is sometimes called as protein blotting or immunoblotting.

It is a widely used analytical technique in the fields of molecular biology, immunogenetics, and other biochemistry disciplines.

Western blotting technique is also used in the field of medica diagnostics. i.e., in the analysis of various kinds of diseases.

Apart from the limitations of western blotting, it is more helpful now a days. Hence, The discovery of Western blotting technique has become boon in the field of science & technology

Regards : Zarva Chaudhary & Chaudhary Moa

Mutations Identification

MUTATION IDENTIFICATION

- Digestion of genomic DNA to DNA fragments.
- Size-separation of the fragments.
- In situ denaturation of the DNA fragments.
- Transfer of denatured DNA fragments into a solid support (nylon or nitrocellulose).
- Hybridization of the immobilized DNA to a labeled probe (DNA, RNA).
- Detection of the bands complementary to the probe (e.g. by autoradiography).

Regards : Zarva Chaudhary & Chaudhary Moazzan

Western Blotting - Procedure

PROCEDURE

3- Transfer can be done in wet or semi-dry conditions. Semi-dry transfer is generally faster. Wet transfer is recommended for large proteins, >100 kD.

4- Blocking is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blotting membrane.

Regards : Zarva Chaudhary & Chaudhary Moazzam

Western Blotting - Procedure

PROCEDURE

1- The choice of extraction method depends primarily on the sample and whether the analysis is targeting all the proteins in a cell or only a component from a particular subcellular fraction.

2- Samples are loaded into separate wells. A protein marker is also loaded. The separated protein mixtures are transferred to a solid support for further analysis.

Regards : Zarva Chaudhary & Chaudhary Moazz

Limitations in western blotting

very delicate and time consuming process. A minute imbalance at any level of the procedure can skew the results of the entire process.

Incorrect labeling of the protein can happen due to the reaction of secondary antibody.

Cause erroneous in bands or no bands due to insufficient transfer.

Well trained technicians are required for this technique.

Primary antibody availability is crucial.

It is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein is possible

Regards : Zarva Chaudhary & Chaudhary Moaz

Mutations Identification

MUTATION IDENTIFICATION

- Mutations can be identified by Southern blotting.

Regards : Zarva Chaudhary & Chaudhary Moazzam

Essentials of Genetics

- Analysis Of IgG Fractions purified from human plasma.
- Diagnosis of HIV by ELISA, involves the western blotting technique.
- Western blotting technique is also used to Detect Some Form Of Lyme Disease.
- Western blotting technique is used in Definitive Test For BSE which is commonly know as Mad cow disease.
- Confirmatory Test For Hepatitis-B involves western blotting technique.
- Western blotting test is used in the Analysis Of Biomarkers such as hormones, growth factors & cytokines.
- This technique is also employed in The Gene Expression Studies

Regards : Zarva Chaudhary & Chaudhary Mo

BIO 203 Quiz

BIO203:Quiz 2

Question # 1 of 10 (Start time: 02:54:20 AM, 13 August 2021)

---------- is sometimes called as protein blotting or immunoblotting

Select the correct option

- ○ Eastern blotting
- ○ Northern blotting
- ○ Southern blotting
- ● Western blotting

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

Type here to search

BIO203:Quiz 2

Question # 2 of 10 (Start time: 02:54:59 AM, 13 August 2021)

Northern blot is used in many ways while studying RNA, for example

Select the correct option

- ○ Study of gene expression in eukaryotic cells
- ○ To measure the amount & size of RNAs transcribed from eukaryotic genes
- ● All of above
- ○ To estimate the abundance of RNAs

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

Type here to search

BIO203:Quiz 2

Question # 3 of 10 (Start time: 02:55:57 AM, 13 August 2021)

During western blotting, transfer can be done in

Select the correct option

- Dry or semi-wet conditions
- Wet or semi-dry conditions

Answer solved by A
Contact: 0305-471
Email: maniamaa

- Dry or semi-dry conditions
- Semi wet or semi-dry conditions

Type here to search

BIO203:Quiz 2

Question # 4 of 10 (Start time: 02:57:10 AM, 13 August 2021)

Western blotting technique is used to

Select the correct option

- Determine the molecular weight of a protein
- Detect the presence of a specific protein
- Measure relative amounts (quantitation) of the protein
- All of above

Answer solved by Amaan Kha
Contact: 0305-4716616
Email: maniamaan2@gmail.c

BIO203:Quiz 2

Question # 9 of 10 (Start time: 03:01:36 AM, 13 August 2021)

Labeled material to detect complementary region in the gene or DNA is called as.

Select the correct option

- ○ Blocking agents
- ○ DNTPs
- ○ Biotin
- ● Probe

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

Type here to search

BIO203:Quiz 2

Question # 10 of 10 (Start time: 03:02:02 AM, 13 August 2021)

The most common antibody label used in Western blots is ----------

Select the correct option

- ○ Biotin
- ● HRP
- ○ Alexa Fluor
- ○ None

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

Type here to search

BIO203:Quiz 2

Question # 7 of 10 (Start time: 03:00:39 AM, 13 August 2021)

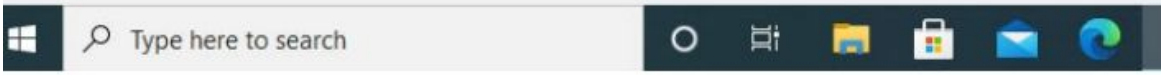
In Western blotting technique, incorrect labeling of the protein can happen due to the reaction of ---------- antibody

Select the correct option

- ○ Primary
- ● Secondary

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

- ○ Tertiary
- ○ None of above



BIO203:Quiz 2

Question # 8 of 10 (Start time: 03:01:10 AM, 13 August 2021)

Western blotting test is used in the analysis of Biomarkers such as--------

Select the correct option

- ○ Cytokines
- ○ Hormones
- ● All

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

- ○ Growth factors

BIO203:Quiz 2

Question # 5 of 10 (Start time: 02:58:37 AM, 13 August 2021)

For the detection and characterization of viral proteins, monoclonal and polyclonal antibody----------technique is used.

Select the correct option

- ○ Southern blotting
- ● Western blotting

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

- ○ Northern blotting
- ○ Eastern blotting

Type here to search

BIO203:Quiz 2

Question # 6 of 10 (Start time: 02:59:22 AM, 13 August 2021)

Systematic steps of Western blotting protocol are

Select the correct option

- ○ 1-Sample preparation,2- Blotting (or transfer),3- Gel Electrophoresis,4- Blocking,5- Detection ,6- Antibody probing
- ○ 1-Antibody probing,2- Gel Electrophoresis,3- Blocking,4- Blotting (or transfer),5- Sample preparation,6- Detection
- ● 1-Sample preparation,2- Gel Electrophoresis,3- Blotting (or transfer),4- Blocking,5- Antibody probing,6- Detection
- ○ 1-Sample preparation,2- Blocking,3- Blotting (or transfer),4- Gel Electrophoresis, 5- Antibody probing,6- Detection

Type here to search

BIO203:Quiz 2

Question # 1 of 10 (Start time: 03:04:25 AM, 13 August 2021)

Diagnosis of HIV by ELISA, involves the ----------blotting technique.

Select the correct option

- ○ Northern
- ● Western

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

- ○ Southern
- ○ Eastern

Type here to search

BIO203:Quiz 2

Question # 2 of 10 (Start time: 03:04:54 AM, 13 August 2021)

During western blotting the choice of extraction method depends primarily on the ----------

Select the correct option

- ○ Sample
- ● All

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

- ○ Whether the analysis is targeting only a subcellular fraction
- ○ Whether the analysis is targeting all the proteins

Type here to search

Question # 3 of 10 (Start time: 03:05:36 AM, 13 August 2021)

Western blotting technique is also used to detect

Select the correct option

●	All
○	Mad cow disease
○	Lyme Disease
○	Hepatitis-B

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

Type here to search

BIO203:Quiz 2

Question # 4 of 10 (Start time: 03:06:11 AM, 13 August 2021)

During western blotting, transfer can be done in

Select the correct option

○	Semi wet or semi-dry conditions
●	Wet or semi-dry conditions
○	Dry or semi-dry conditions
○	Dry or semi-wet conditions

Answer solved by Amaan Kha
Contact: 0305-4716616
Email: maniamaan2@gmail.c

BIO203:Quiz 2

Question # 5 of 10 (Start time: 03:06:26 AM, 13 August 2021)

Which is not true for Radioactive Probe - p32

Select the correct option

- Sensitive
- Radioactive waste disposal regulations should not be followed
- Hazardous
- Relatively cheap

Answer solve
Contact: 030
Email: mania

Type here to search

BIO203:Quiz 2

Question # 6 of 10 (Start time: 03:07:00 AM, 13 August 2021)

Systematic steps of Western blotting protocol are

Select the correct option

- 1-Sample preparation,2- Blotting (or transfer),3- Gel Electrophoresis,4- Blocking,5- Detection ,6- Antibody probing
- 1-Antibody probing,2- Gel Electrophoresis,3- Blocking,4- Blotting (or transfer),5- Sample preparation,6- Det
- 1-Sample preparation,2- Blocking,3- Blotting (or transfer),4- Gel Electrophoresis, 5- Antibody probing,6- De
- 1-Sample preparation,2- Gel Electrophoresis,3- Blotting (or transfer),4- Blocking,5- Antibody probing,6- Det

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VU Medical Zo

IO203:Quiz 2

uestion # 9 of 10 (Start time: 03:08:08 AM, 13 August 2021)

----------is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blott

elect the correct option

- Blocking
- Antibody probing
- Gel Electrophoresis
- Blotting or transfer

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com



IO203:Quiz 2

uestion # 10 of 10 (Start time: 03:08:41 AM, 13 August 2021)

Western blotting technique is simply a way to identify unknown ----------on a polyacrylamide gel

elect the correct option

- Proteins
- Carbohydrate
- DNA
- RNA

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com



BIO203:Quiz 2

Question # 7 of 10 (Start time: 03:07:38 AM, 13 August 2021)

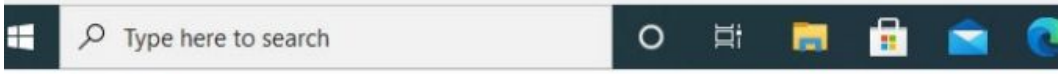
Labeled material to detect complementary region in the gene or DNA is called as.

Select the correct option

- (●) Probe

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

- ( ) Blocking agents
- ( ) DNTPs
- ( ) Biotin



BIO203:Quiz 2

Question # 8 of 10 (Start time: 03:07:54 AM, 13 August 2021)

The most common antibody label used in Western blots is ----------

Select the correct option

- ( ) None
- (●) HRP

Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com

- ( ) Alexa Fluor
- ( ) Biotin

